

Solutions to Problem Solving Class 4: IP

Computer Communications – DE
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1. A: 128
B: 16368
C: 2096896
2. A: 16777216
B: 65536
C: 256
3. a) C
b) D
c) A
d) B
e) E
4. a) A, 114.0.0.0
b) B, 171.34.0.0
c) C, 192.8.56.0
d) C, 205.23.67.0
5. a) Net-id: 130.235. Host-id: 185.49.
b) Net-id: 130.235.128. Host-id: 60.247.
6. a) /24
b) /8
c) /19
d) /20
7. a) 123.56.77.32-123.56.77.39
b) 200.17.21.128-200.17.21.159
c) 17.34.16.0-17.34.17.255
d) 180.34.64.64-180.34.64.67
8. a) 82EB120A → 130.235.18.10

- b) $82EB127F \rightarrow 130.235.18.127$
 - c) 4
 - d) $5 \cdot 4 \text{ bytes} = 20 \text{ bytes}$
 - e) $3E_{16} = 3 \cdot 16 + 14 = 62_{10} \text{ bytes}$
 - f) $80_{16} = 8 \cdot 16 = 128_{10}$
9. Check the protocol field (should be 01).
10. a) IP may be carried by another protocol that restricts the frame size.
 b) Check the flags+fragmentation field; 000|0000000000000 means that we may fragment the message if necessary, that this is the last or only fragment, and also that the offset is zero. Thus this packet is not fragmented.
 c) The TCP header contains the port number. When a packet is fragmented, the TCP header ends up in the first packet only and the other packets contain only TCP data, which confuse the router.
11. a) Frame
 1: 001 $\rightarrow$ can be fragmented, is not the last fragment.
 2: 001 $\rightarrow$ can be fragmented, is not the last fragment.
 3: 000 $\rightarrow$ can be fragmented, last or only fragment.
 b) The three fragments stem from the same original datagram.
12. IPv6 uses the concept of *Next Header*. A header option is inserted as an extra fixed-size header (extension header). This speeds up the routing process.
13. a) 2340:1ABC:119A:A000::0
 b) 0:AA::119:A231
 c) 2340::119A:A001:0
 d) 0:0:0:2340::0
14. a) 0000:0000:0000:0000:0000:0000:0000:0000
 b) 0000:00AA:0000:0000:0000:0000:0000:0000
 c) 0000:1234:0000:0000:0000:0000:0000:0003
 d) 0123:0000:0000:0000:0000:0000:0001:0002
15. Theoretically: An IPv6 address is $8 \cdot 2 \cdot 8 = 128$ bits. Thus there are 2^{128} address possibilities. An IPv4 address is only $8 \cdot 4 = 32$ bits, which gives 2^{32} address possibilities. The difference is $2^{128} - 2^{32}$ (huge!).
16. The IP header is 20 bytes long if we omit the optional padding field. The checksum field occurs as the 11:th and the 12:th byte of the IP header.

Frame 1: $DA4F_{16}$ is indeed the resulting checksum. The frame is correct.

Frame 2: $5A4F_{16}=0101101001001111_2$ is the computed checksum. The frame is not correct.

17. The resulting IP header is:

4500013E00AB400080067341C0A80254C0A80229